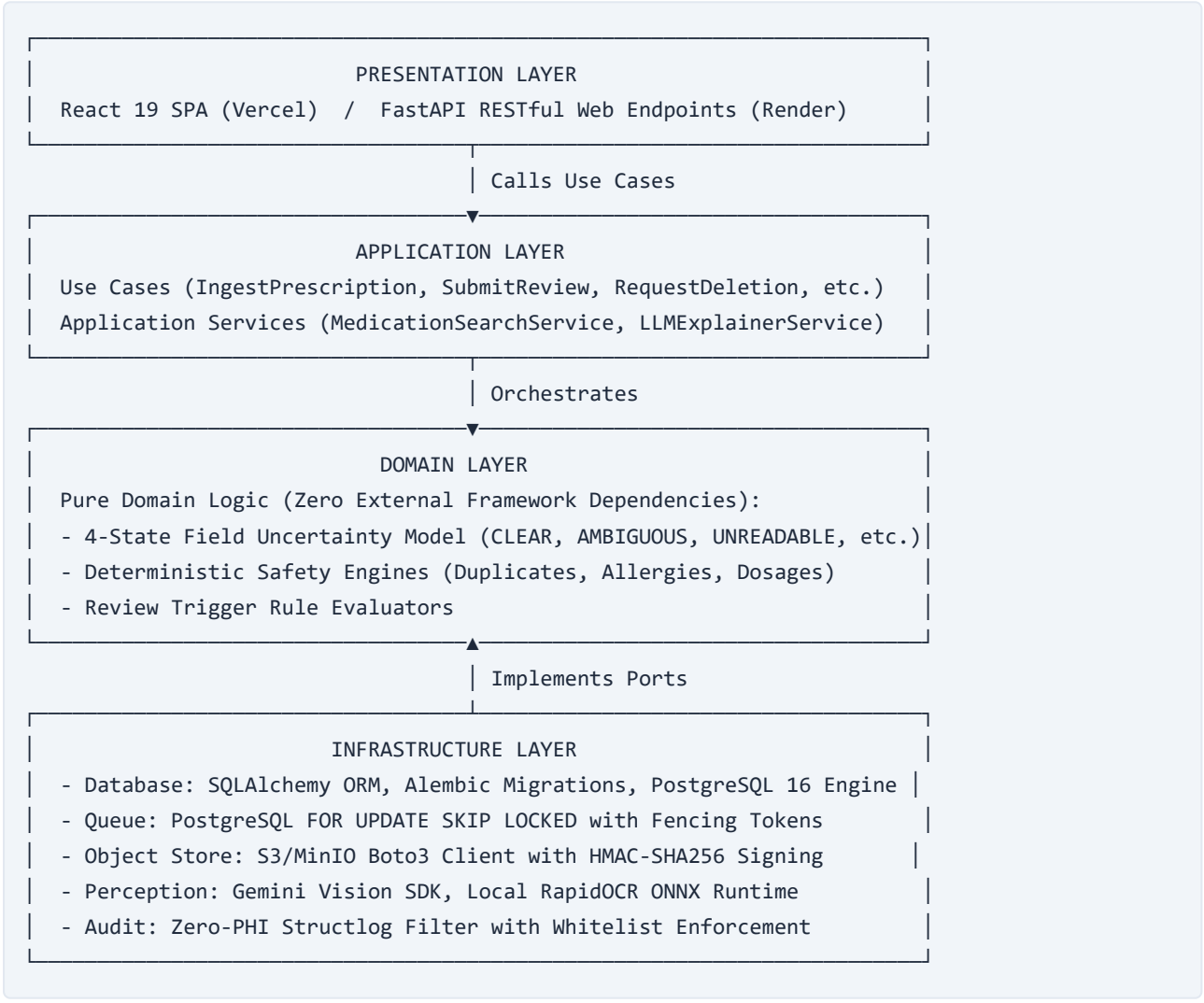
Prescripto AI 2.0 — System Architecture Dossier

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1. Architectural Philosophy and Style

Prescripto AI 2.0 is engineered as a **Hexagonal / Clean Architecture (Modular Monolith)** coupled to an **Asynchronous Fenced Worker Subsystem**:



Architectural Guarantees

1. **Separation of Concerns**: The Domain layer (`backend/prescripto/domain/`) has zero dependencies on FastAPI, SQLAlchemy, or Boto3. It can be compiled and verified completely in isolation.
2. **Deterministic Safety Decoupling**: Probabilistic vision models (Gemini / PaddleOCR) are strictly confined to perception. No neural model makes final decisions on drug-drug interactions, duplicate therapies, or dosage violations.

3. **Audit Immutability**: Once an analysis is finalized by a clinician sign-off (`REVIEWED_COMPLETE`), the database locks the record against any subsequent programmatic modification.

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2. End-to-End System Component Architecture

```
flowchart TD
    subgraph Client Tier
        Browser[Clinician / Patient Web Browser]
        VercelEdge[Vercel CDN Edge Network]
        Browser -->|Fetches Static Assets| VercelEdge
    end

    subgraph API Gateway & Presentation Tier
        RenderLB[Render Cloud Reverse Proxy]
        FastAPIApp[FastAPI Modular Monolith :8000]
        CORS[CORS Middleware: allow_origin_regex]
        Audit[Zero-PHI Structlog Middleware]

        Browser -->|HTTPS REST API Calls| RenderLB
        RenderLB -->|Proxy Headers: X-Forwarded-Proto| FastAPIApp
        FastAPIApp --> CORS
        FastAPIApp --> Audit
    end

    subgraph Data & Storage Tier
        Postgres[(PostgreSQL 16 Managed DB)]
        S3Store[(MinIO / AWS S3 Object Store)]
    end

    subgraph Asynchronous Worker Tier
        WorkerEngine[Fenced Background Worker Engine]
        Heartbeat[Heartbeat Thread: Lease Renewer]
        StageEngine[8-Stage Sequential Pipeline Engine]

        WorkerEngine -->|Claim Job: FOR UPDATE SKIP LOCKED| Postgres
        WorkerEngine --> Heartbeat
        WorkerEngine --> StageEngine
    end

    subgraph Perception & AI Services
        GeminiCloud[Google Gemini Vision API: 3.5-Flash / 3.1-Flash-Lite]
        LocalRapid[Local RapidOCR ONNX Runtime on CPU]
    end

    FastAPIApp -->|Persists Document & Job| Postgres
    FastAPIApp -->|Streams Image Bytes| S3Store
    StageEngine -->|Stage 1: Fetch Scan| S3Store
    StageEngine -->|Stage 3: Vision Perception| GeminiCloud
    GeminiCloud -.->|On Quota / Timeout Failover| LocalRapid
    StageEngine -->|Stage 6-8: Persist Results| Postgres
```

3. Asynchronous Execution & Generation Fencing Model

To prevent duplicate job processing and race conditions without adding Redis or Celery overhead, Prescripto AI 2.0 leverages PostgreSQL's native row-level concurrency primitives:

```
sequenceDiagram
    autonumber
    participant W1 as Worker Node 1
    participant DB as PostgreSQL (analysis_jobs)
    participant W2 as Worker Node 2
    participant HB as Heartbeat Thread

    Note over DB: Job 101 status = QUEUED

    W1->>DB: SELECT * FROM analysis_jobs WHERE status='QUEUED'<br/>ORDER BY created_at FOR UPDAT
    DB-->>W1: Grants Row Lock on Job 101
    W1->>DB: UPDATE analysis_jobs SET status='RUNNING', worker_id='worker-1',<br/>lease_token='l
    W1->>HB: Launch Heartbeat Thread (interval: 30s)

    Note over W1: Worker 1 executes Stage 1 - 4

    W2->>DB: SELECT * FROM analysis_jobs WHERE status='QUEUED'<br/>FOR UPDATE SKIP LOCKED LIMIT
    Note over W2: Skips Job 101 (Locked by Worker 1)
    DB-->>W2: Returns 0 rows (No available jobs)

    loop Every 30 Seconds
        HB->>DB: UPDATE analysis_jobs SET heartbeat_at=NOW(),<br/>expires_at=NOW() + 10m WHERE i
        DB-->>HB: Lease Successfully Renewed
    end

    W1->>DB: Commit Stage 8 & Mark Analysis Status = REQUIRES_REVIEW
    W1->>HB: Stop Heartbeat Thread
    W1->>DB: UPDATE analysis_jobs SET status='COMPLETED'
```

The Worker Fencing Guarantee (`WorkerFencedError`)

If Worker 1 experiences a long GC pause or network partition exceeding `lease_seconds` (10 minutes), Worker 2 can reclaim the job. When Worker 1 recovers and attempts to commit its next stage result, the query validates:

```
UPDATE analysis_jobs
SET heartbeat_at = NOW()
WHERE id = :job_id AND lease_token = :my_token;
```

If `rowcount == 0`, Worker 1 raises `WorkerFencedError`, immediately rolls back its transaction, and ceases execution, guaranteeing that stale workers never corrupt pipeline state.

---

4. Uncertainty Propagation Architecture

The pipeline enforces an explicit mathematical contract across all 8 stages:

```
stateDiagram-v2
    [*] --> Ingested
    Ingested --> QualityChecked

    QualityChecked --> ClearField: Image Sharp (Var > 100) & Conf >= 0.85
    QualityChecked --> AmbiguousField: Image Borderline or Conf 0.50..0.84
    QualityChecked --> UnreadableField: Image Blurry (Var < 100) or Conf < 0.50

    state ClearField {
        [*] --> HighConfidenceExtraction
        HighConfidenceExtraction --> ExactFormularyMatch: Matches 1 Verified CDSCO Item
        ExactFormularyMatch --> ResolvedMedication: ResolutionStatus = RESOLVED
    }

    state AmbiguousField {
        [*] --> LowConfidenceExtraction
        LowConfidenceExtraction --> AmbiguousMatch: Matches Multiple Candidates
        AmbiguousMatch --> RequiresReview: ResolutionStatus = AMBIGUOUS
    }

    state UnreadableField {
        [*] --> MaskedStroke
        MaskedStroke --> ExcludedFromSafetyRules: Value = Masked
        ExcludedFromSafetyRules --> RequiresReview: ResolutionStatus = UNRESOLVED
    }

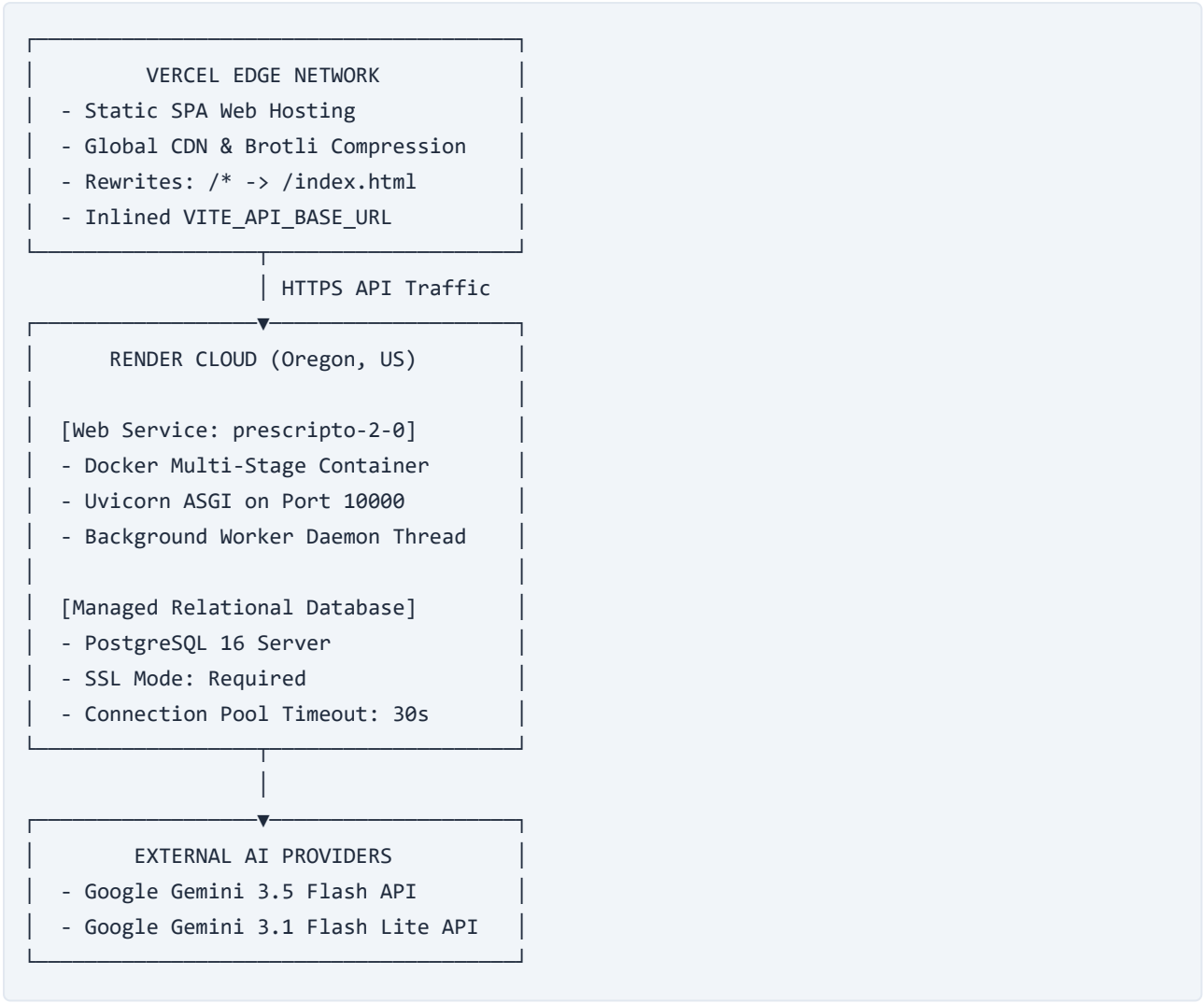
    ResolvedMedication --> SafetyScreening
    SafetyScreening --> ReportAssembled
    RequiresReview --> ReportAssembled

    ReportAssembled --> ClinicianSignOff: Status = REQUIRES_REVIEW
    ClinicianSignOff --> Approved: Clinician Confirms / Overrides
    Approved --> [*]
```

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5. Deployment Topology

The production architecture utilizes a resilient multi-cloud split topology:


VERCEL EDGE NETWORK
- Static SPA Web Hosting
- Global CDN & Brotli Compression
- Rewrites: /* -> /index.html
- Inlined VITE_API_BASE_URL
HTTPS API Traffic
RENDER CLOUD (Oregon, US)
[Web Service: prescripto-2-0]
- Docker Multi-Stage Container
- Uvicorn ASGI on Port 10000
- Background Worker Daemon Thread
[Managed Relational Database]
- PostgreSQL 16 Server
- SSL Mode: Required
- Connection Pool Timeout: 30s
EXTERNAL AI PROVIDERS
- Google Gemini 3.5 Flash API
- Google Gemini 3.1 Flash Lite API